

```
Enter the investment amount (greater than 0 but less than 50,000): 1500
Enter the annual interest rate (greater than 0 but less than 15): 6
Enter the investment duration in years (greater than 0): 10
Year 1: $1,592.52
Year 2: $1,690.73
Year 3: $1,795.01
Year 4: $1,905.73
Year 5: $2,023.29
Year 6: $2,148.09
Year 8: $2,421.25
Year 9: $2,570.60
Year 10: $2,729.15
```

```
Year 5: $2,023.29
Year 6: $2,148.09
Year 8: $2,421.25
Year 9: $2,570.60
Year 10: $2,729.15
Final Investment Details:
Total investment after 10 years: $2,729.15
Initial investment: $1500
Annual interest rate: 6%
Monthly compounded total: $2,729.15
```

interest_calculation.py

Function to handle invalid user input for investment amount

```
def get_investment_amount():
```

```
    while True:
```

```
        try:
```

```
            investment = int(input("Enter the investment amount (greater than 0 but less than 50,000): "))
```

```
            if 0 < investment < 50000:
```

```
                return investment
```

```
            else:
```

```
                print("Investment must be between 0 and 50,000. Please try again.")
```

```
        except ValueError:
```

```
            print("Invalid input! Please enter a valid number.")
```

```
# Function to handle invalid user input for interest rate
```

```
def get_interest_rate():
```

```
    while True:
```

```
        try:
```

```
            rate = int(input("Enter the annual interest rate (greater than 0 but less than 15): "))
```

```
            if 0 < rate < 15:
```

```
                return rate
```

```
            else:
```

```
                print("Interest rate must be between 0 and 15. Please try again.")
```

```
        except ValueError:
```

```
            print("Invalid input! Please enter a valid number.")
```

```
# Function to handle invalid user input for investment duration
```

```
def get_investment_duration():
```

```
    while True:
```

```
        try:
```

```
            years = int(input("Enter the investment duration in years (greater than 0): "))
```

```
            if years > 0:
```

```
                return years
```

```
            else:
```

```
                print("Duration must be greater than 0. Please try again.")
```

```
        except ValueError:
```

```
            print("Invalid input! Please enter a valid number.")
```

```
# Main logic for calculating compounded investment
```

```
def calculate_investment():  
    # Get inputs from the user  
    investment = get_investment_amount()  
    interest_rate = get_interest_rate()  
    duration_years = get_investment_duration()  
  
    # Convert years to months  
    months = duration_years * 12  
  
    # Convert annual interest rate to monthly decimal rate  
    monthly_rate = (interest_rate / 12) / 100  
  
    # Initialize the total investment  
    total = investment  
  
    # Loop through each month to calculate compound interest  
    for month in range(1, months + 1):  
        # Calculate interest for the current month  
        interest = total * monthly_rate  
        # Add the interest to the total amount  
        total += round(interest, 2)  
  
    # Check if it's the end of a year to print the results  
    if month % 12 == 0:  
        year = month // 12  
        print(f"Year {year}: ${round(total, 2):,.2f}")
```

```
# Final output after all compounding is complete

print("\nFinal Investment Details:")

print(f"Total investment after {duration_years} years: ${round(total, 2):,.2f}")

print(f"Initial investment: ${investment}")

print(f"Annual interest rate: {interest_rate}%")

print(f"Monthly compounded total: ${round(total, 2):,.2f}")


if __name__ == "__main__":
    calculate_investment()

"Completed by, Manuel Vargas"
```